Databricks ML Associate

High-Yield Next 100 (Set 2)

Practice Exam Deck

How to Use This Deck

100 more high-yield questions (Q1–Q100), complementing Set 1, on topics that commonly appear on the Databricks Machine Learning Associate exam.

Each question slide has four options (A–D); the next slide gives the correct answer and why the others are wrong.

Topic order: platform, compute, Delta and Spark; MLflow; model registry; Feature Engineering; data processing; tuning; Spark ML; metrics and concepts; deployment, jobs, and monitoring.

Exam Domains Covered

Databricks Machine Learning: runtime, AutoML, MLflow, Unity Catalog registry, Feature Engineering client.

Data Processing: EDA, imputation, encoding, Spark ML transformers, pandas API on Spark and UDFs.

Model Development: Hyperopt, CrossValidator, metrics, bias/variance, leakage.

Model Deployment: batch, streaming, real-time serving, monitoring.

Question 1

What does the auto-termination setting on a cluster do?

A. Restarts the cluster nightly
B. Deletes the notebooks attached
C. Shuts the cluster down after a set period of inactivity
D. Stops a job when it fails

Answer 1

Correct: C — it saves cost on idle compute.

Wrong:

A: not what it does.

B: notebooks are not deleted.

D: that is a job setting.

Question 2

What is the purpose of a cluster policy?

A. To schedule notebooks
B. To register models
C. To store secrets
D. To restrict and standardize the configuration users can choose

Answer 2

Correct: D — admins use policies to control size, runtime, and cost.

Wrong:

A: that is Workflows.

B: that is the model registry.

C: that is secret scopes.

Question 3

What does cluster autoscaling do?

A. Changes the driver runtime version

B. Adds or removes workers between a minimum and maximum based on load

C. Increases the dataset size

D. Scales the model endpoint to zero

Answer 3

Correct: B — it matches compute to the workload.

Wrong:

A: not what it does.

C: not what it does.

D: that is serving.

Question 4

Which magic command runs another notebook inline so its functions and variables are available?

A. `%fs`

B. `%md`

C. `%run`

D. `%sh`

Answer 4

Correct: C — `%run` includes the other notebook in the current one.

Wrong:

A: runs file system commands.

B: renders markdown.

D: runs shell commands.

Question 5

After `%pip install` changes a package already imported, what should you do?

A. Run `VACUUM`

B. Run `dbutils.library.restartPython()`

C. Run `df.cache()`

D. Run `mlflow.end_run()`

Answer 5

Correct: B — restarting Python picks up the new package version.

Wrong:

A: a Delta command.

C: unrelated.

D: ends a run only.

Question 6

Which Delta Lake command performs an upsert?

A. `OPTIMIZE`

B. `DESCRIBE HISTORY`

C. `MERGE INTO`

D. `INSERT OVERWRITE DIRECTORY`

Answer 6

Correct: C — it updates matching rows and inserts new ones.

Wrong:

A: compacts files.

B: shows the table log.

D: writes files to a directory.

Question 7

What does `OPTIMIZE` do on a Delta table?

A. Deletes all old versions
B. Encrypts the table
C. Rolls back to an earlier version
D. Compacts many small files into larger ones

Answer 7

Correct: D — it improves read performance.

Wrong:

A: that is `VACUUM`.

B: not what it does.

C: that is `RESTORE`.

Question 8

What does `VACUUM` do on a Delta table?

A. Clones the table
B. Removes data files no longer referenced and older than the retention period
C. Adds a column
D. Compacts small files

Answer 8

Correct: **B** — this frees storage but limits time travel.

Wrong:

A: that is `CLONE`.

C: that is `ALTER TABLE`.

D: that is `OPTIMIZE`.

Question 9

Which command shows the version history of a Delta table?

A. `DESCRIBE TABLE EXTENDED` only

B. `SHOW TABLES`

C. `DESCRIBE HISTORY table_name`

D. `EXPLAIN table_name`

Answer 9

Correct: C — it lists versions, timestamps, and operations.

Wrong:

A: shows metadata, not versions.

B: lists tables.

D: shows a query plan.

Question 10

Which command restores a Delta table to an earlier version?

A. `UNDO TABLE t 5`

B. `OPTIMIZE t VERSION 5`

C. `VACUUM t VERSION 5`

D. `RESTORE TABLE t TO VERSION AS OF 5`

Answer 10

Correct: D — it rolls the table back to that version.

Wrong:

A: not valid.

B: OPTIMIZE compacts files.

C: VACUUM deletes files.

Question 11

What does `ZORDER BY` do in `OPTIMIZE table ZORDER BY (col)`?

A. Encrypts the column

B. Co-locates related values of the column in the same files to speed up filters

C. Sorts the output of a query only

D. Deletes duplicate rows

Answer 11

Correct: **B** — it improves data skipping.

Wrong:

A: not what it does.

C: it changes file layout.

D: not what it does.

Question 12

Which write saves a DataFrame as a managed Delta table?

A. `df.cache("t")`

B. `df.toTable("t")`

C. `df.save_delta("t")`

D. `df.write.mode("overwrite").saveAsTable("catalog.schema.t")`

Answer 12

Correct: D — saveAsTable registers the table in the catalog.

Wrong:

A: caching does not persist a table.

B: not an API.

C: not an API.

Question 13

What is the difference between `repartition(n)` and `coalesce(n)`?

A. They are identical

B. `coalesce` always increases partitions

C. `coalesce` reduces partitions without a full shuffle; `repartition` does a full shuffle

D. `repartition` only decreases partitions

Answer 13

Correct: C — coalesce is cheaper when decreasing partitions.

Wrong:

A: they differ in shuffling.

B: it only decreases.

D: it can increase or decrease.

Question 14

When is a broadcast join beneficial?

A. When both tables are huge
B. When using only streaming data
C. When one table is small enough to copy to every executor
D. When there is no join key

Answer 14

Correct: C — it avoids shuffling the large table.

Wrong:

A: broadcasting would not fit.

B: unrelated.

D: unrelated.

Question 15

What is a shuffle in Spark?

A. Randomly ordering the notebook cells
B. Copying a table to another catalog
C. Encrypting data in memory
D. Redistribution of data across partitions and workers, often costly

Answer 15

Correct: D — joins and groupBy usually cause shuffles.

Wrong:

A: unrelated.

B: unrelated.

C: unrelated.

Question 16

Which Spark call triggers computation?

A. `df.count()`

B. `df.select("a")`

C. `df.filter("a > 1")`

D. `df.withColumn("b", df.a)`

Answer 16

Correct: A — count is an action.

Wrong:

B: a transformation.

C: a transformation.

D: a transformation.

Question 17

Which privilege lets a user read a Unity Catalog table (along with USE CATALOG and USE SCHEMA)?

A. `SELECT`

B. `WRITE VOLUME`

C. `EXECUTE`

D. `CREATE MODEL`

Answer 17

Correct: A — SELECT grants read access to table data.

Wrong:

B: applies to volumes.

C: is for functions and models.

D: creates models in a schema.

Question 18

What does Unity Catalog lineage provide?

A. Faster clusters
B. Automatic tracking of how tables, columns, and models relate to upstream data
C. Automatic hyperparameter tuning
D. Free storage

Answer 18

Correct: B — it helps with impact analysis and governance.

Wrong:

A: not provided.

C: not provided.

D: not provided.

Question 19

Which call logs several hyperparameters at once?

A. `mlflow.log_artifacts_params()`

B. `mlflow.log_params({"max_depth": 5, "n_estimators": 100})`

C. `mlflow.set_params()`

D. `mlflow.log_param_dict()`

Answer 19

Correct: **B** — log_params takes a dictionary.

Wrong:

A: not an API.

C: not an API.

D: not an API.

Question 20

Which call logs several metrics at once?

A. `mlflow.log_score()`

B. `mlflow.log_metric_dict()`

C. `mlflow.set_metrics()`

D. `mlflow.log_metrics({"rmse": 3.1, "r2": 0.8})`

Answer 20

Correct: D — log_metrics takes a dictionary.

Wrong:

A: not an API.

B: not an API.

C: not an API.

Question 21

What is `mlflow.set_tag("team", "risk")` best used for?

A. Adding searchable string metadata to a run
B. Saving model weights
C. Logging a numeric metric
D. Changing the experiment

Answer 21

Correct: A — tags label and organize runs.

Wrong:

B: that is `log_model`.

C: that is `log_metric`.

D: that is `set_experiment`.

Question 22

Which call returns the currently active run object?

A. `mlflow.current_model()`

B. `mlflow.active_run()`

C. `mlflow.run_id()`

D. `mlflow.get_active_experiment()`

Answer 22

Correct: **B** — it returns the active run or None.

Wrong:

A: not an API.

C: not an API.

D: not the run.

Question 23

How do you read the run ID inside an active run?

A. `mlflow.get_id()`

B. `mlflow.run_id`

C. `mlflow.active_run().metrics`

D. `mlflow.active_run().info.run_id`

Answer 23

Correct: D — run_id is stored in the run info.

Wrong:

A: not an API.

B: not an attribute.

C: holds metrics.

Question 24

What does `mlflow.sklearn.load_model(uri)` return?

A. A Spark DataFrame

B. The native scikit-learn model object

C. A run object

D. A PyFuncModel always

Answer 24

Correct: B — flavor-specific loaders return the native object.

Wrong:

A: not what it returns.

C: not what it returns.

D: that is `mlflow.pyfunc.load_model`.

Question 25

What does `mlflow.pyfunc.load_model(uri)` return?

A. A Hyperopt Trials object

B. A generic model with a `predict()` method

C. The raw training DataFrame

D. A feature table

Answer 25

Correct: **B** — it works across flavors.

Wrong:

A: not what it returns.

C: not what it returns.

D: not what it returns.

Question 26

Which file in a logged MLflow model folder describes its flavors and metadata?

A. `cluster.json`

B. `README.md`

C. `train.csv`

D. `MLmodel`

Answer 26

Correct: D — MLmodel lists flavors, signature, and more.

Wrong:

A: not generated.

B: not generated.

C: not generated.

Question 27

Which files record the model's Python dependencies?

A. `requirements.txt` and `conda.yaml`

B. `MLmodel` only

C. `notebook.ipynb`

D. `signature.json`

Answer 27

Correct: A — they let the environment be recreated for serving.

Wrong:

B: does not list all packages.

C: not a dependency file.

D: not generated.

Question 28

How do you implement a custom model with preprocessing and postprocessing logic in MLflow?

A. Subclass `mlflow.pyfunc.PythonModel` and implement `predict()`

B. Subclass `mlflow.Run`

C. Override `mlflow.start_run`

D. Write a Delta trigger

Answer 28

Correct: A — a custom pyfunc wraps arbitrary logic.

Wrong:

B: not for models.

C: not for models.

D: not for models.

Question 29

Which call evaluates a logged model with default metrics and plots?

A. `mlflow.validate_run()`

B. `mlflow.score()`

C. `mlflow.test_model()`

D. `mlflow.evaluate(...)`

Answer 29

Correct: D — it computes metrics and artifacts for a model and dataset.

Wrong:

A: not an API.

B: not an API.

C: not an API.

Question 30

How can you disable MLflow autologging?

A. `mlflow.stop_autolog()`

B. `mlflow.autolog(disable=True)`

C. `mlflow.end_run(autolog=False)`

D. Delete the experiment

Answer 30

Correct: B — the disable flag turns autologging off.

Wrong:

A: not an API.

C: not a parameter.

D: not needed.

Question 31

Where does MLflow Tracking on Databricks store run metadata by default?

A. In a Git repository
B. Only on the cluster's local disk
C. In the feature store
D. In the workspace-managed tracking server for the experiment

Answer 31

Correct: D — Databricks hosts tracking for you.

Wrong:

A: not used for runs.

B: would be lost on termination.

C: not used for runs.

Question 32

Which client method lists versions of a registered model?

A. `MlflowClient().search_model_versions("name='catalog.schema.model'")`

B. `mlflow.search_runs()`

C. `MlflowClient().list_versions_df()`

D. `mlflow.get_versions()`

Answer 32

Correct: A — it filters model versions by name.

Wrong:

B: searches runs.

C: not an API.

D: not an API.

Question 33

Which tag or name argument sets a friendly name for a run?

A. `mlflow.end_run("baseline")`

B. `mlflow.start_run(label="baseline")`

C. `mlflow.start_run(run_name="baseline")`

D. `mlflow.log_param("name")`

Answer 33

Correct: C — run_name labels the run.

Wrong:

A: ends a run.

B: not an argument.

D: logs a param.

Question 34

What does `mlflow.end_run()` do?

A. Deletes the experiment
B. Stops the cluster
C. Ends the active run
D. Registers the model

Answer 34

Correct: C — a run ends when the context exits or when end_run is called.

Wrong:

A: not what it does.

B: not what it does.

D: not what it does.

Question 35

In Unity Catalog, what must exist before a model can be registered?

A. A serving endpoint

B. A feature table

C. A catalog and schema where the user has CREATE MODEL

D. An AutoML experiment

Answer 35

Correct: C — models live inside schemas.

Wrong:

A: created later.

B: not required.

D: not required.

Question 36

How can you attach metadata such as validation status to a model version?

A. Rename the experiment
B. Change the Delta schema
C. Use model version tags, for example `set_model_version_tag`
D. Edit the cluster policy

Answer 36

Correct: C — tags store key-value metadata.

Wrong:

A: does not tag the version.

B: unrelated.

D: unrelated.

Question 37

What is a common practice before moving the Champion alias to a new version?

A. Validate the new version against checks and a baseline
B. Skip tests for speed
C. Delete the old version immediately
D. Retrain without logging

Answer 37

Correct: A — validation reduces production risk.

Wrong:

B: risky.

C: prevents rollback.

D: loses traceability.

Question 38

Which URI loads a specific numbered version of a UC registered model?

A. `runs:/model/3`

B. `models:/3/catalog.schema.model`

C. `versions:/model@3`

D. `models:/catalog.schema.model/3`

Answer 38

Correct: D — the version number follows the model name.

Wrong:

A: not a model version URI.

B: wrong order.

C: not valid.

Question 39

Which statement about aliases is true?

A. One alias points to one version at a time within a model

B. An alias can point to many versions at once

C. Aliases are only for the legacy registry

D. Aliases delete old versions

Answer 39

Correct: A — reassigning moves the alias.

Wrong:

B: it points to one.

C: they are the UC mechanism.

D: they do not.

Question 40

A model in Unity Catalog was trained on features from a UC feature table. What does UC lineage show?

A. The upstream feature table and data sources linked to the model

B. Only the run duration

C. Nothing, models have no lineage

D. Only the cluster size

Answer 40

Correct: A — lineage connects data assets and models.

Wrong:

B: not lineage.

C: UC tracks model lineage.

D: not lineage.

Question 41

Which call reads a feature table as a Spark DataFrame?

A. `fe.read_table(name="catalog.schema.features")`

B. `fe.create_table()`

C. `fe.load_features()`

D. `fe.drop_table()`

Answer 41

Correct: A — read_table returns the table data.

Wrong:

B: creates a table.

C: not an API.

D: deletes a table.

Question 42

Which call deletes a feature table?

A. `fe.write_table(mode="delete")`

B. `fe.remove()`

C. `fe.drop_table(name=...)`

D. `fe.delete_features()`

Answer 42

Correct: C — drop_table removes the feature table.

Wrong:

A: not a mode.

B: not an API.

D: not an API.

Question 43

What does `exclude_columns` do in `create_training_set`?

A. Excludes tables from lineage

B. Excludes failed runs

C. Removes listed columns from the resulting training DataFrame

D. Excludes rows with nulls

Answer 43

Correct: C — use it to drop keys or other non-feature columns.

Wrong:

A: unrelated.

B: unrelated.

D: it drops columns.

Question 44

Which problem does a feature store help reduce?

A. Cluster costs only

B. Label noise

C. Training-serving skew caused by inconsistent feature code

D. Model overfitting

Answer 44

Correct: C — one definition of features is used everywhere.

Wrong:

A: not its purpose.

B: not directly solved.

D: not directly solved.

Question 45

A feature is computed from request-time input such as the current transaction amount. What helps?

A. An on-demand feature function computed at inference

B. Dropping the feature

C. Storing it in a notebook cell

D. A nightly batch feature only

Answer 45

Correct: A — feature functions compute values that cannot be precomputed.

Wrong:

B: loses information.

C: not governed.

D: would be stale.

Question 46

What is a feature table in Unity Catalog, technically?

A. A serving endpoint
B. A CSV file in DBFS
C. A pandas DataFrame in memory
D. A Delta table with a primary key constraint

Answer 46

Correct: D — UC feature tables are Delta tables with keys.

Wrong:

A: not a table.

B: not a feature table.

C: not persisted.

Question 47

Which parameter provides the join key column when defining a `FeatureLookup`?

A. `lookup_key`

B. `join_col_name`

C. `label`

D. `primary_keys`

Answer 47

Correct: A — it names the key column in the label DataFrame.

Wrong:

B: not the parameter.

C: used in the training set.

D: used when creating tables.

Question 48

Which `FeatureLookup` parameter enables point-in-time joins?

A. `feature_names`

B. `timestamp_lookup_key`

C. `version_as_of`

D. `time_travel=True`

Answer 48

Correct: B — it names the timestamp column in the label data.

Wrong:

A: lists features only.

C: not a parameter.

D: not a parameter.

Question 49

What is the difference between normalization (min-max) and standardization?

A. Min-max rescales to a fixed range; standardization centers to mean 0 and unit variance
B. They are identical
C. Standardization maps values to 0 or 1
D. Min-max centers at the median

Answer 49

Correct: A — different scaling approaches for different needs.

Wrong:

B: they differ.

C: that is binarization.

D: not true.

Question 50

Which Spark ML class standardizes features?

A. `StandardScaler`

B. `VectorAssembler`

C. `StringIndexer`

D. `Tokenizer`

Answer 50

Correct: A — it scales to unit variance and optionally centers.

Wrong:

B: combines columns.

C: indexes categories.

D: splits text.

Question 51

Which Spark ML class rescales features to a given range such as 0 to 1?

A. `Bucketizer`

B. `PCA`

C. `Imputer`

D. `MinMaxScaler`

Answer 51

Correct: D — it does min-max scaling.

Wrong:

A: bins continuous values.

B: reduces dimensions.

C: fills missing values.

Question 52

Which Spark ML class bins a continuous column using specified split points?

A. `StringIndexer`

B. `OneHotEncoder`

C. `StandardScaler`

D. `Bucketizer`

Answer 52

Correct: D — it maps values into buckets.

Wrong:

A: indexes strings.

B: encodes indexes.

C: scales values.

Question 53

What does PCA do?

A. Balances classes

B. Projects data to fewer uncorrelated components that capture most variance

C. Tunes hyperparameters

D. Fills missing values

Answer 53

Correct: B — it reduces dimensionality.

Wrong:

A: not its purpose.

C: not its purpose.

D: not its purpose.

Question 54

What problem does multicollinearity cause for a linear model?

A. It improves interpretability
B. It removes the intercept
C. It makes training impossible on Spark
D. Unstable and hard-to-interpret coefficients

Answer 54

Correct: D — correlated predictors make coefficients unreliable.

Wrong:

A: it harms it.

B: not true.

C: not true.

Question 55

Which measure is commonly used to detect multicollinearity?

A. ROC AUC

B. Silhouette score

C. Log loss

D. Variance inflation factor (VIF)

Answer 55

Correct: D — a high VIF suggests collinearity.

Wrong:

A: classification metric.

B: clustering metric.

C: classification loss.

Question 56

A nominal categorical feature such as color has no natural order. Which encoding is appropriate for a linear model?

A. Dropping the column always

B. Squaring the values

C. Ordinal integer encoding

D. One-hot encoding

Answer 56

Correct: D — it avoids imposing a false order.

Wrong:

A: loses information.

B: not an encoding.

C: implies a false order.

Question 57

Why can target encoding cause leakage?

A. It always lowers accuracy

B. It removes all labels

C. It requires a GPU

D. It uses label information that can leak into training features if not computed carefully

Answer 57

Correct: D — compute it within folds or on training data only.

Wrong:

A: not true.

B: not true.

C: not true.

Question 58

A feature has 40 percent missing values. Which step is reasonable?

A. Investigate why, then impute or add a missing indicator, or drop if uninformative
B. Ignore it silently
C. Always drop all such rows
D. Always fill with zero

Answer 58

Correct: A — choose based on cause and importance.

Wrong:

B: can break models.

C: may lose too much data.

D: can distort the data.

Question 59

Which operation removes duplicate rows in Spark?

A. `df.dropDuplicates()`

B. `df.dedupe()`

C. `df.removeRows()`

D. `df.unique()`

Answer 59

Correct: A — it removes duplicates, optionally on a subset of columns.

Wrong:

B: not an API.

C: not an API.

D: not an API.

Question 60

Which Spark call removes rows containing nulls?

A. `df.fillna()`

B. `df.dropDuplicates()`

C. `df.dropna()` (`df.na.drop()`)

D. `df.nullify()`

Answer 60

Correct: C — it drops rows with missing values.

Wrong:

A: fills nulls.

B: removes duplicates.

D: not an API.

Question 61

Which call fills nulls with a constant in a Spark DataFrame?

A. `df.fillna(0)` (`df.na.fill(0)`)

B. `df.replaceNull(0)`

C. `df.dropna(0)`

D. `df.coalesce(0)`

Answer 61

Correct: A — it replaces nulls with the given value.

Wrong:

B: not an API.

C: drops rows.

D: changes partitions.

Question 62

Which pandas UDF type lets you apply Python logic to batches of a column?

A. A Delta trigger
B. A Series-to-Series pandas UDF
C. A row-at-a-time Python UDF
D. A SQL view

Answer 62

Correct: **B** — it receives and returns a pandas Series per batch.

Wrong:

A: not a UDF.

C: is not vectorized.

D: not a UDF.

Question 63

What should a Hyperopt objective function return so `fmin` can record status and loss?

A. A list of models

B. Nothing

C. A dict such as `{"loss": value, "status": STATUS_OK}`

D. A pandas DataFrame

Answer 63

Correct: C — fmin reads the loss and status keys.

Wrong:

A: not accepted.

B: a loss is required.

D: not accepted.

Question 64

In `hp.loguniform("lr", low, high)`, what are `low` and `high`?

A. The mean and std

B. The natural logs of the range limits

C. The actual range limits

D. The number of trials

Answer 64

Correct: B — so use np.log(1e-5) and np.log(1e-1) for 1e-5 to 1e-1.

Wrong:

A: not their meaning.

C: they are logs.

D: not their meaning.

Question 65

What does `max_evals` control in `fmin`?

A. The number of CV folds

B. The number of workers

C. The tree depth

D. The maximum number of trials to evaluate

Answer 65

Correct: D — it bounds the search budget.

Wrong:

A: set in the objective.

B: that is parallelism.

C: a model hyperparameter.

Question 66

Which function converts the index results from `fmin` back to the actual parameter values?

A. `hp.resolve()`

B. `space_eval(space, best)`

C. `tpe.decode()`

D. `eval_space()`

Answer 66

Correct: B — it maps hp.choice indexes to values.

Wrong:

A: not an API.

C: not an API.

D: not an API.

Question 67

Which Hyperopt object stores all trial results for later inspection?

A. `STATUS_OK`

B. `Trials` (or `SparkTrials`)

C. `tpe.suggest`

D. `hp.choice`

Answer 67

Correct: B — pass it to fmin to keep trial info.

Wrong:

A: a status constant.

C: an algorithm.

D: defines a space.

Question 68

Why should the objective function evaluate on validation data, not training data?

A. Training loss rewards overfit hyperparameters
B. Training data is too small
C. Hyperopt forbids training data
D. Validation data is faster

Answer 68

Correct: A — validation estimates generalization.

Wrong:

B: not the reason.

C: not a rule.

D: not the reason.

Question 69

What does `numFolds` set in Spark ML `CrossValidator`?

A. The number of workers
B. The number of folds
C. The number of features
D. The number of trees

Answer 69

Correct: B — the data is split into that many folds.

Wrong:

A: not set here.

C: not set here.

D: a model parameter.

Question 70

How do you build a parameter grid in Spark ML?

A. `GridSearch.build()`

B. `ParamGridBuilder().addGrid(param, values).build()`

C. `Hyperopt.grid()`

D. `CrossValidator.grid()`

Answer 70

Correct: B — the builder creates the combinations.

Wrong:

A: not Spark ML.

C: not an API.

D: not an API.

Question 71

After fitting a `CrossValidator`, how do you access the best model?

A. `cv_model.best_run`

B. `cv_model.top`

C. `cv_model.bestModel`

D. `cv_model.winner`

Answer 71

Correct: C — the fitted CrossValidatorModel exposes bestModel.

Wrong:

A: not an attribute.

B: not an attribute.

D: not an attribute.

Question 72

What does `avgMetrics` contain in a `CrossValidatorModel`?

A. The feature importances
B. The list of columns
C. The average cross-validation metric for each parameter combination
D. The training time per fold

Answer 72

Correct: C — use it to compare grid points.

Wrong:

A: not contained.

B: not contained.

D: not contained.

Question 73

Which Spark ML class implements logistic regression?

A. `LinearRegression`

B. `LogisticRegression`

C. `PCA`

D. `KMeans`

Answer 73

Correct: B — it trains binary and multiclass classifiers.

Wrong:

A: predicts continuous targets.

C: reduces dimensions.

D: is clustering.

Question 74

In Spark ML regularization, what does `elasticNetParam=1.0` mean?

A. A 50/50 mix

B. No regularization

C. Pure L2 penalty

D. Pure L1 (Lasso) penalty

Answer 74

Correct: D — 0 is L2 and 1 is L1.

Wrong:

A: that is 0.5.

B: that is `regParam=0`.

C: that is 0.

Question 75

Which parameter sets the regularization strength in Spark ML linear models?

A. `featuresCol`

B. `numTrees`

C. `regParam`

D. `maxBins`

Answer 75

Correct: C — larger values mean stronger regularization.

Wrong:

A: column name.

B: forest size.

D: tree discretization.

Question 76

Which parameter controls maximum tree depth in Spark ML decision trees?

A. `maxDepth`

B. `maxIter`

C. `numFolds`

D. `regParam`

Answer 76

Correct: A — deeper trees can overfit.

Wrong:

B: iterations.

C: CV folds.

D: regularization.

Question 77

Which Spark ML estimator performs clustering?

A. `LogisticRegression`
B. `DecisionTreeClassifier`
C. `KMeans`
D. `GBTRegressor`

Answer 77

Correct: C — it groups rows into k clusters.

Wrong:

A: supervised.

B: supervised.

D: supervised.

Question 78

Which evaluator should you use to compute silhouette score for clustering in Spark ML?

A. `BinaryClassificationEvaluator`

B. `RegressionEvaluator`

C. `MulticlassClassificationEvaluator`

D. `ClusteringEvaluator`

Answer 78

Correct: D — its default metric is silhouette.

Wrong:

A: binary classification.

B: regression metrics.

C: classification metrics.

Question 79

Which Spark ML algorithm builds recommendations with collaborative filtering?

A. `PCA`

B. `ALS`

C. `LinearSVC`

D. `KMeans`

Answer 79

Correct: B — alternating least squares for user-item ratings.

Wrong:

A: dimensionality reduction.

C: classification.

D: clustering.

Question 80

Which statement about fitting preprocessing in a pipeline is correct?

A. Fit separately on each split

B. Fit on training data only and apply the fitted transformers to validation and test data

C. Fit on the full dataset first

D. Fit on test data only

Answer 80

Correct: B — this prevents leakage.

Wrong:

A: creates inconsistent features.

C: leaks information.

D: defeats the purpose.

Question 81

What is precision?

A. TP / (TP + FP)

B. TP / (TP + FN)

C. (TP + TN) / total

D. TN / (TN + FP)

Answer 81

Correct: A — the share of predicted positives that are correct.

Wrong:

B: that is recall.

C: that is accuracy.

D: that is specificity.

Question 82

What is recall?

A. TP / (TP + FP)
B. (TP + TN) / total
C. FP / (FP + TN)
D. TP / (TP + FN)

Answer 82

Correct: D — the share of actual positives that are found.

Wrong:

A: that is precision.

B: that is accuracy.

C: that is false positive rate.

Question 83

What does the F1 score combine?

A. The arithmetic mean of accuracy and AUC
B. The harmonic mean of precision and recall
C. The sum of TP and TN
D. The maximum of precision and recall

Answer 83

Correct: B — it balances both.

Wrong:

A: not its definition.

C: not its definition.

D: not its definition.

Question 84

Which is true about MSE versus MAE units?

A. MAE is squared

B. Both are unitless

C. MSE is in squared units; MAE is in the target's units

D. MSE equals RMSE

Answer 84

Correct: C — RMSE returns to the original units.

Wrong:

A: MSE is squared.

B: not true.

D: RMSE is the square root of MSE.

Question 85

Why can MAPE be a poor metric?

A. It ignores errors

B. It cannot be computed in Spark

C. It is undefined or unstable when actual values are zero or near zero

D. It only works for classification

Answer 85

Correct: C — it divides by the actual value.

Wrong:

A: not true.

B: not the reason.

D: it is for regression.

Question 86

What does adjusted R2 do that R2 does not?

A. Predicts probabilities

B. Penalizes adding predictors that do not improve the model

C. Ignores the number of features

D. Replaces the test set

Answer 86

Correct: B — plain R2 never decreases when features are added.

Wrong:

A: unrelated.

C: R2 does that.

D: unrelated.

Question 87

What does log loss penalize heavily?

A. Missing values
B. Large feature counts
C. Confident but wrong probability predictions
D. Slow training

Answer 87

Correct: C — it uses the predicted probability of the true class.

Wrong:

A: not a loss factor.

B: not a loss factor.

D: not a loss factor.

Question 88

What is the difference between a parameter and a hyperparameter?

A. Hyperparameters are learned by gradient descent
B. They are the same
C. Parameters are set by the user
D. Parameters are learned from data; hyperparameters are set before training

Answer 88

Correct: D — for example coefficients versus tree depth.

Wrong:

A: they are set beforehand.

B: they differ.

C: that is hyperparameters.

Question 89

What are the roles of validation and test sets?

A. Validation guides tuning; test gives a final unbiased estimate
B. They are always identical
C. Test guides tuning; validation is final
D. Both are used for training

Answer 89

Correct: A — do not tune on the test set.

Wrong:

B: they must be separate.

C: reversed.

D: they are held out.

Question 90

What is the curse of dimensionality?

A. Training always becomes faster
B. Models always improve
C. Data becomes sparse and models harder to fit as the number of features grows
D. Features become perfectly correlated

Answer 90

Correct: C — more features need far more data.

Wrong:

A: not true.

B: not true.

D: not true.

Question 91

What is the effect of a gradient descent learning rate that is too large?

A. Training always converges faster and better
B. Nothing changes
C. The model underfits by design
D. Training can overshoot and diverge

Answer 91

Correct: D — steps are too big to converge.

Wrong:

A: not guaranteed.

B: the rate matters.

C: not the typical result.

Question 92

What does early stopping do?

A. Stops registering models
B. Stops training when validation performance stops improving
C. Stops reading data
D. Stops the cluster when idle

Answer 92

Correct: **B** — it helps prevent overfitting.

Wrong:

A: not related.

C: not related.

D: that is auto-termination.

Question 93

What happens with scale-to-zero on a Model Serving endpoint?

A. The registry is cleared

B. Predictions are cached forever

C. Compute scales down when there is no traffic, and the next request may see a cold start

D. The model is deleted

Answer 93

Correct: C — it saves cost at the price of latency.

Wrong:

A: not true.

B: not true.

D: the model remains.

Question 94

Which call queries a Databricks serving endpoint from Python using MLflow?

A. `mlflow.query_endpoint()`

B. `mlflow.deployments.get_deploy_client("databricks").predict(endpoint=..., inputs=...)`

C. `spark.predict(endpoint)`

D. `mlflow.serve(endpoint)`

Answer 94

Correct: **B** — the deployments client calls the endpoint.

Wrong:

A: not an API.

C: not an API.

D: not valid.

Question 95

What authentication does a REST call to a Databricks serving endpoint need?

A. A bearer token (personal access token or OAuth token)

B. A Git token

C. No authentication

D. A cluster ID

Answer 95

Correct: A — the request carries an Authorization header.

Wrong:

B: not used.

C: endpoints are secured.

D: not used.

Question 96

In a multi-task Databricks job, how do you make task B run only after task A?

A. Put them in one cell
B. Use two clusters
C. Set task A as a dependency of task B
D. Give them the same name

Answer 96

Correct: C — dependencies define the DAG.

Wrong:

A: not a job feature.

B: not how it works.

D: not how it works.

Question 97

What does repairing a failed job run do?

A. Reruns only the failed and downstream tasks
B. Deletes the job
C. Reruns every task from scratch
D. Resets the cluster policy

Answer 97

Correct: A — successful tasks are not rerun.

Wrong:

B: not what it does.

C: that is a new run.

D: not what it does.

Question 98

How can a downstream task read a value produced by an upstream task?

A. Use `df.cache()`

B. Use task values with `dbutils.jobs.taskValues`

C. Use a global Python variable

D. Use `dbutils.fs.mv`

Answer 98

Correct: B — task values pass small values between tasks.

Wrong:

A: not shared across tasks.

C: tasks run in separate contexts.

D: moves files.

Question 99

Which statistical tests are commonly used to detect drift?

A. VIF and PCA
B. OPTIMIZE and VACUUM
C. ROC AUC and F1
D. Kolmogorov-Smirnov for numeric and chi-square for categorical features

Answer 99

Correct: D — they compare distributions.

Wrong:

A: not drift tests.

B: Delta maintenance.

C: model metrics.

Question 100

A model's online accuracy drops over time though code is unchanged. What is the best response?

A. Ignore it

B. Increase cluster size

C. Delete the model registry

D. Monitor for drift and retrain on fresh data when thresholds are exceeded

Answer 100

Correct: D — production data can change.

Wrong:

A: performance will continue to degrade.

B: does not fix drift.

C: does not help.